

Using MATSim for strategic planning

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Strategy and Long-term Planning

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MATSim User Meeting

Using MATSim to explore
new railway concepts.

Dissolving the clock-face integrated timetable



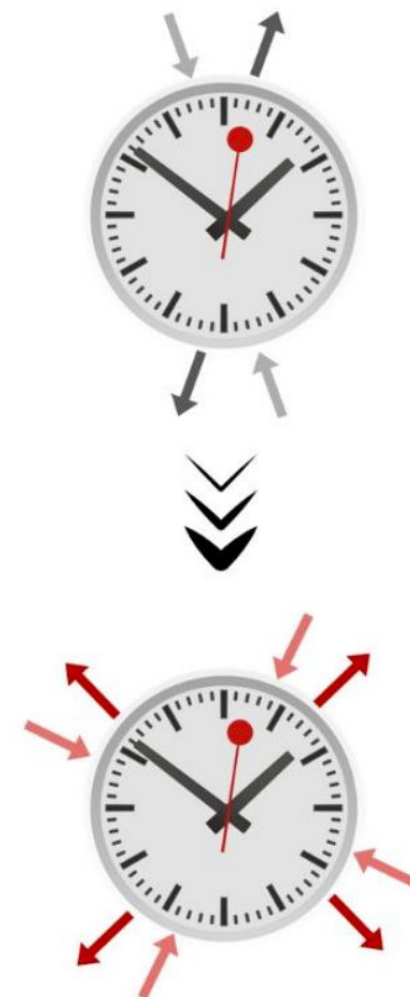
Clock-face Scheduling

1982

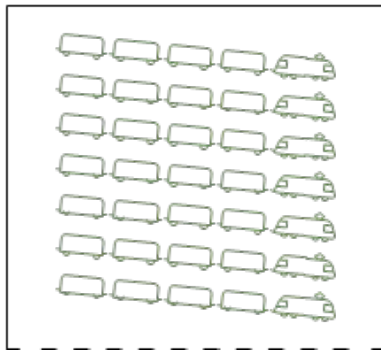
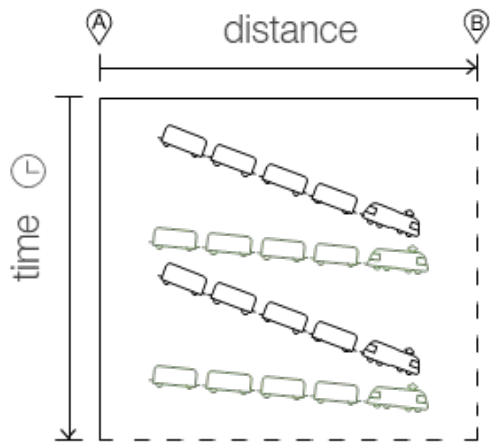


No trains in Zürich HB at 11:45. (Image: D. Eckenstein)

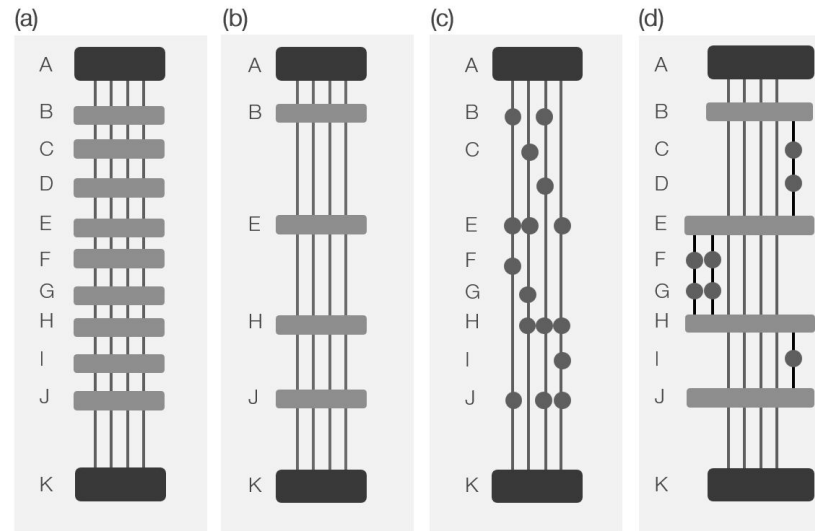
- Today, attractive transfer times are achieved by clock-face integrated timetable.
- At all major nodes, all trains arrive shortly before 00/30 and leave shortly after 00/30.
- New concept: Timetable with dissolved nodes accompanied by higher frequency keep transfer times attractive and use full station capacity.



Harmonization of speeds

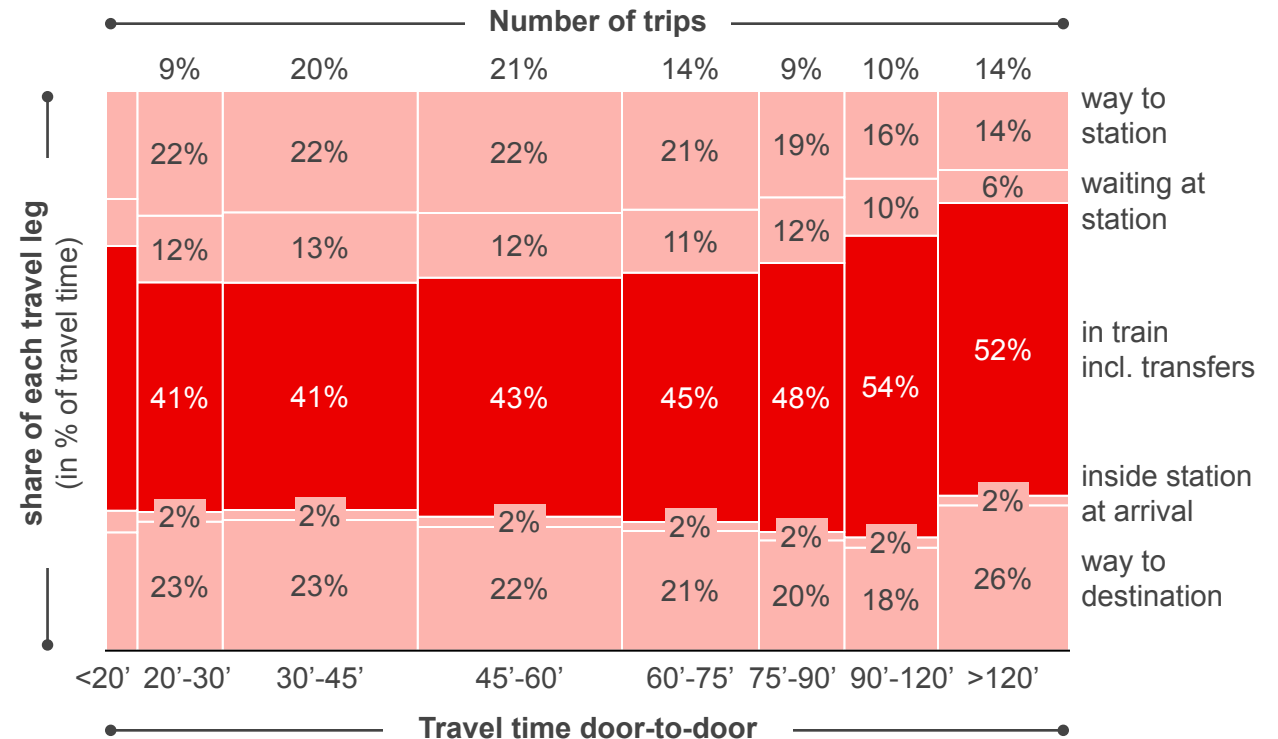
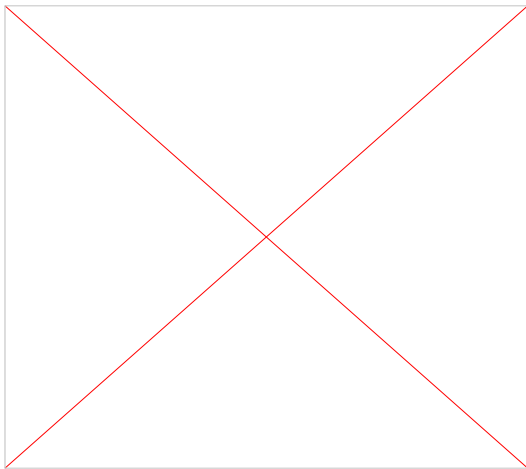


- Line capacity depends on train speed equality on the line:
 - Heterogenous speeds $\square$ low capacity
 - Homogenous speeds $\square$ high capacity
- Average speed depends on $v_{\max}$, but also on operational aspects (stop pattern, crossings/overtaking, ...).
- Various stopping patterns can be used to balance service quality and line capacity



Optimized feeder system

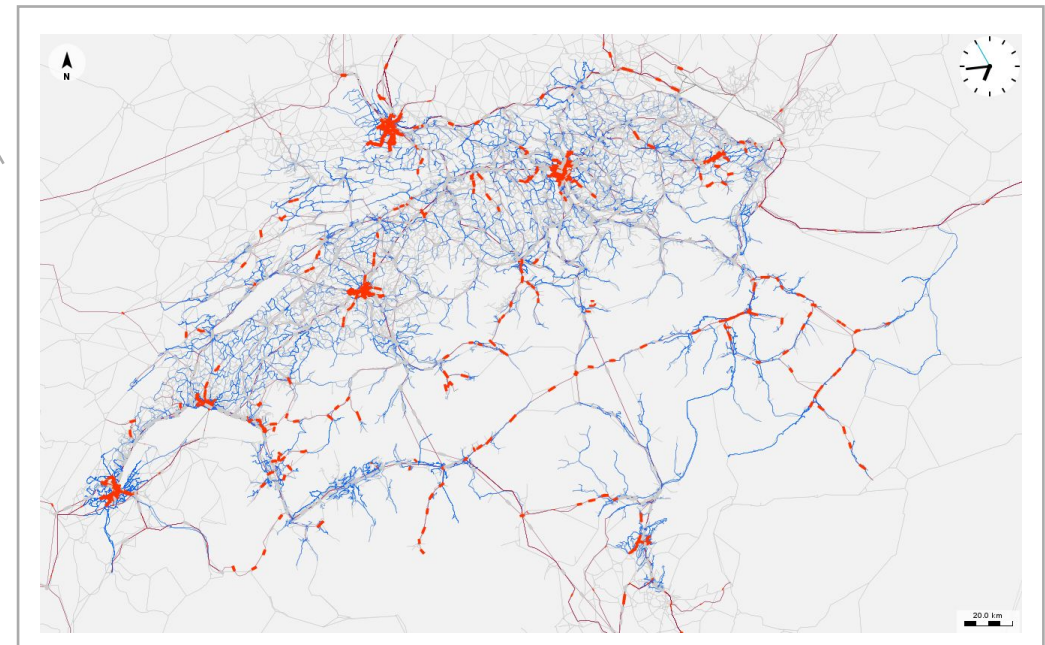
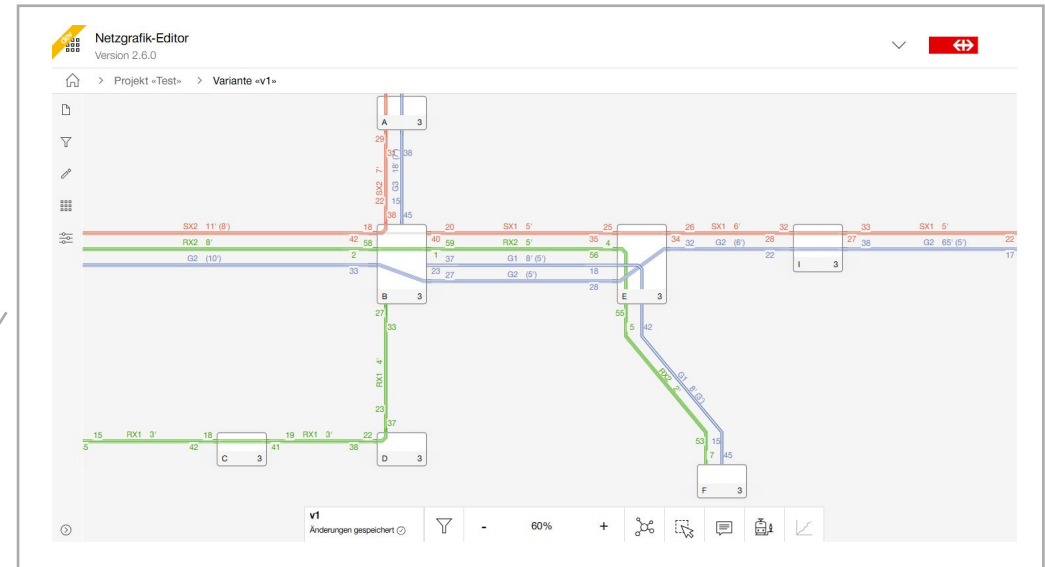
- A good railway system relies on efficient feeder networks: Only **40% to 50%** of the total travel time for railway journeys is spent on the train.
- Autonomous and on-demand feeder systems present opportunities to enhance the accessibility of the railway network.



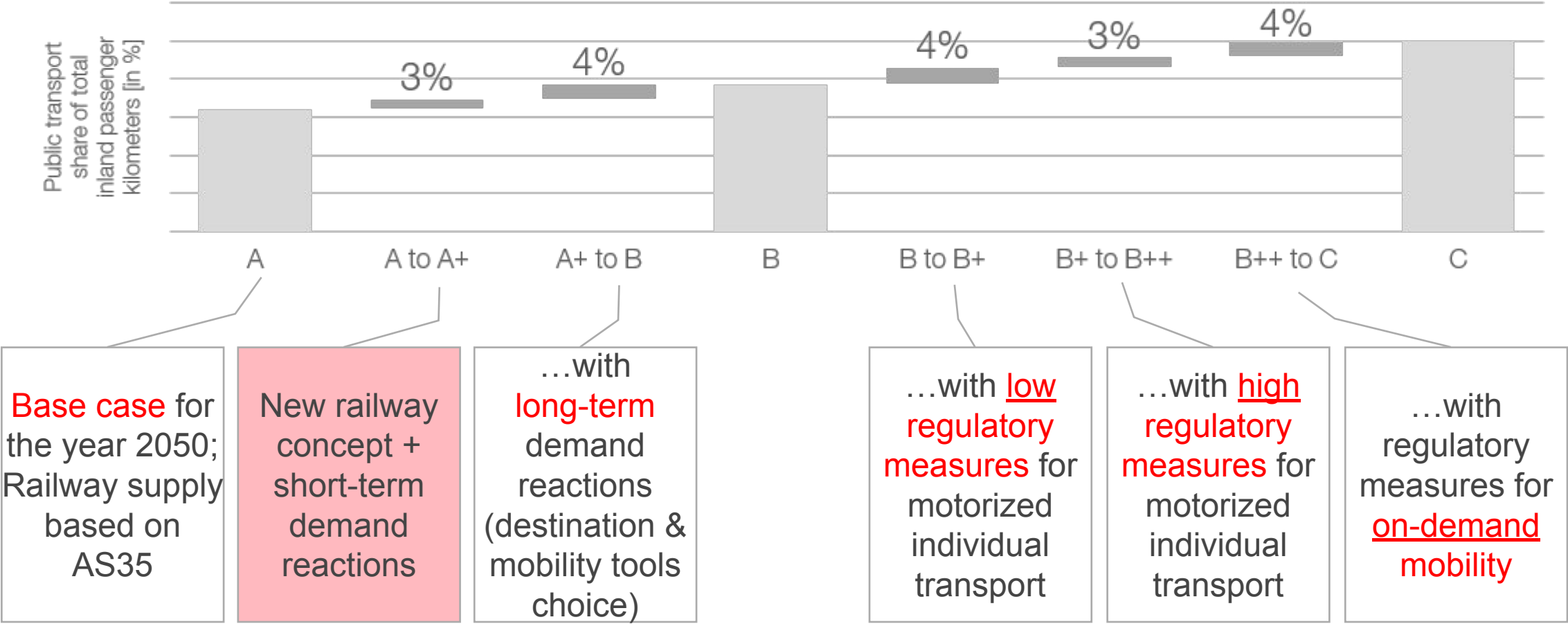
Source: BFS/ARE, Mikrozensus Mobilität und Verkehr (MZMV, 2021)

We used an iterative design-evaluation workflow.

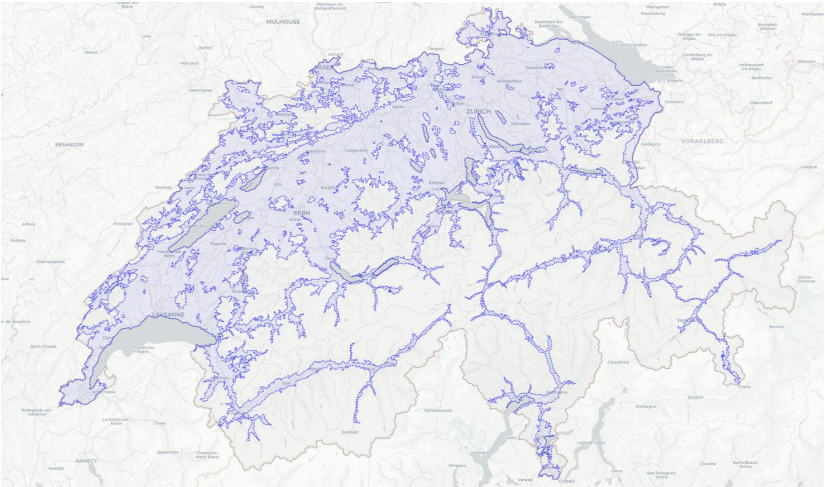
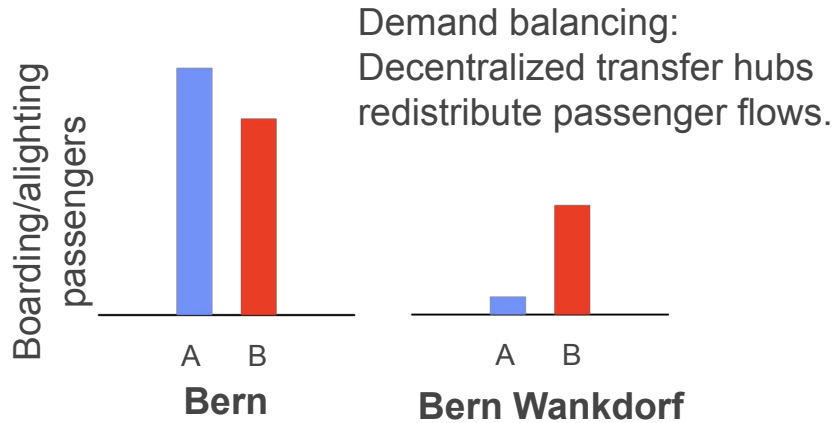
1. The entire **railway supply (AS35)** is **removed** from the existing transport model.
2. A new railway timetable is designed for entire Switzerland using an open source **network graphic editor**.
3. The network plan is converted to the MATSim format using an open-source **network graphic converter**. The new rail supply is added to the model.
4. Depending on the scenario case, **regulatory policies** are implemented.
5. The **simulation** calculates demand responses and provides relevant KPIs.
6. Based on the simulation results, the **network plan is revised** (Goto 2)



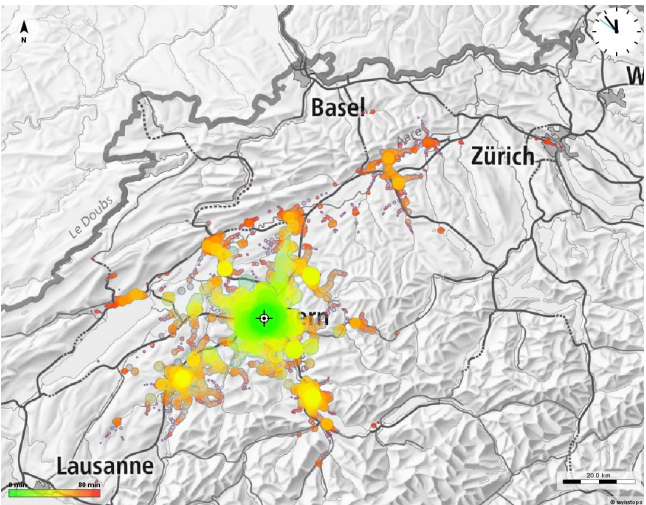
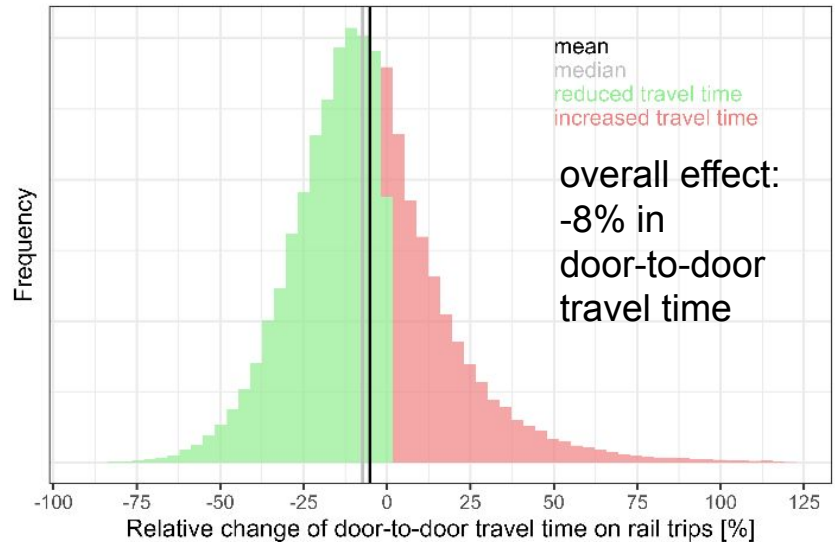
The aggregated modal split effect shows a significant shift towards public transport.



Some further analysis results



96% of the population is located within a 15-minute catchment area of a railway station. With 20% fewer stations it's still 95%.

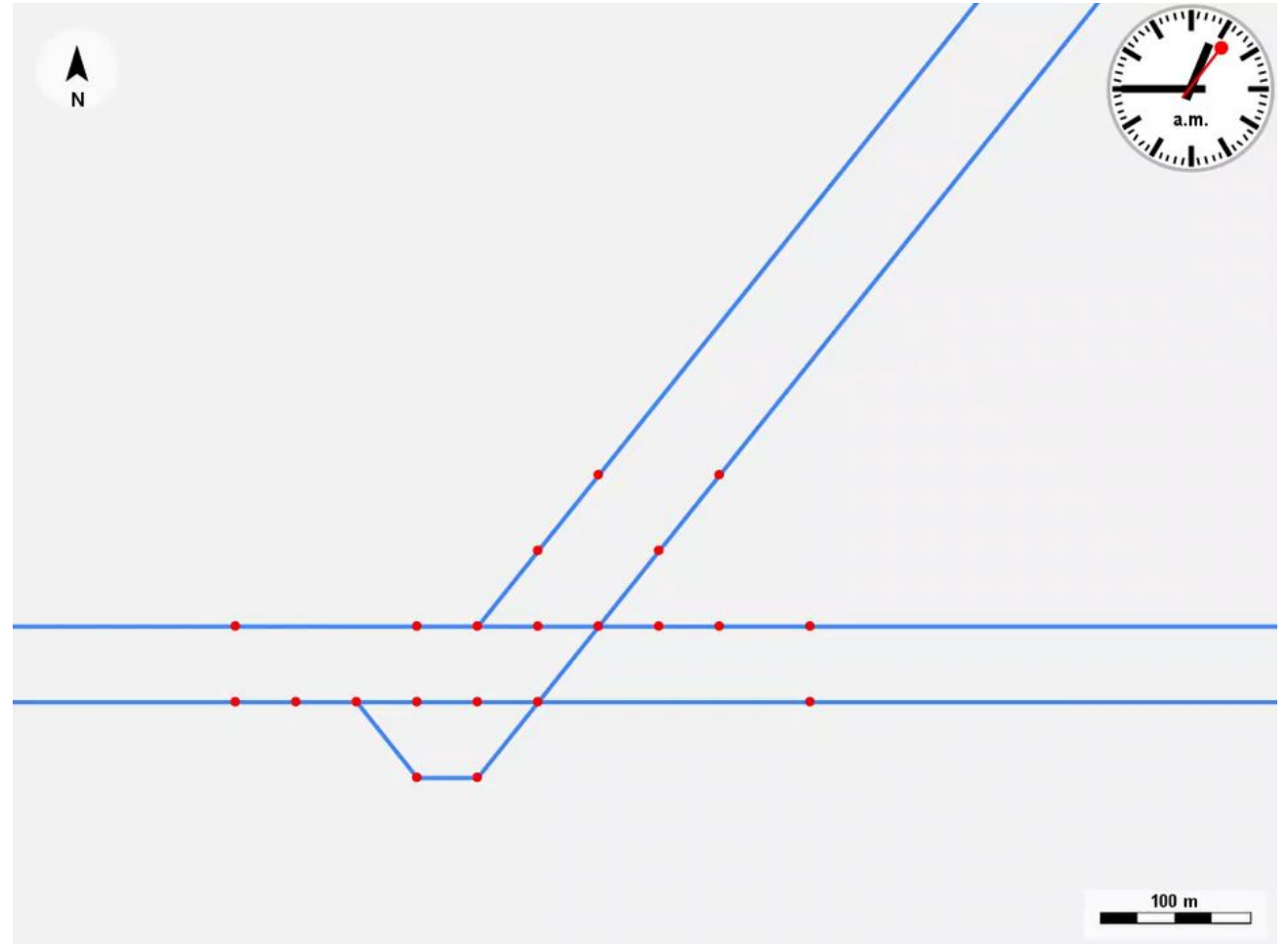


Using MATSim to explore railway operation.

MATSim's railsim contribution



- Fully **compatible** with MATSim's data structure (network, transit schedule, transit vehicles)
- Trains have a **physical length**.
- **Driving dynamics** incorporate acceleration and deceleration.
- Infrastructure is **reserved** based on braking distances.
- Capacity effects are modeled at the level of **resources**.
- **Routing flexibility** allows trains to deviate from predefined network paths.



What is a “reliable capacity” for a given building block and operational schemes?

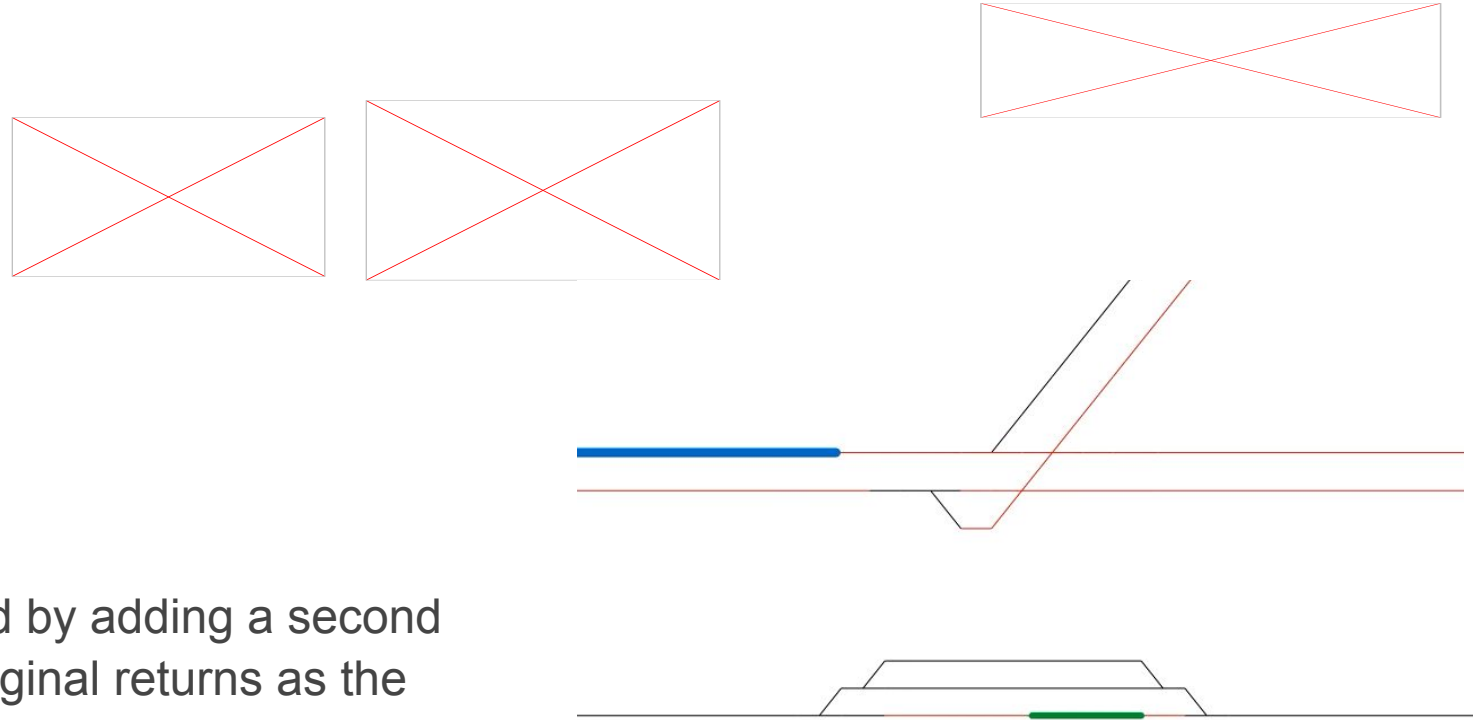
Building blocks: Track topologies

Operational schemes: Product mix, stop pattern, frequency

“Reliable capacity”: The train throughput under varied conditions (randomly sampled schedules)

Main findings:

- Significant capacity gains are achieved by adding a second platform track; a third track shows marginal returns as the bottleneck shifts to the single-track section.
- Grade separation resolves the crossing conflict but shifts the bottleneck to the merging point on the trunk line.

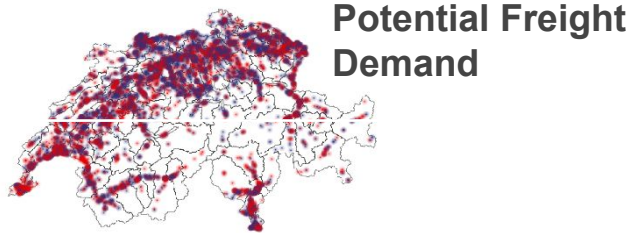


Using MATSim to simulate cargo trains.

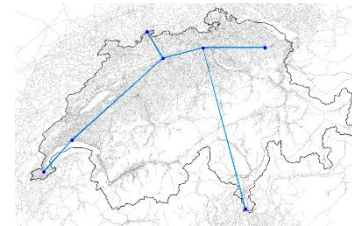
Simulating combined road and rail cargo transport.

Optimization / Experiments

Input

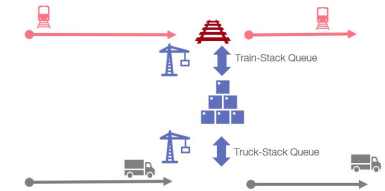


illustrative example

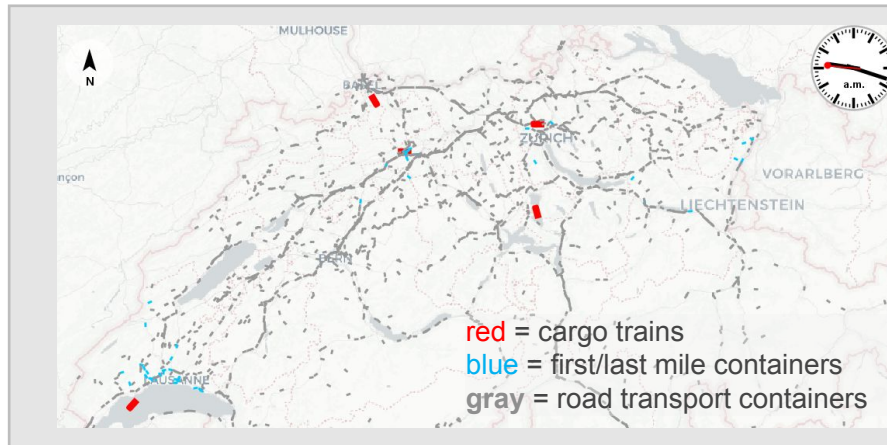


Supply

- Train schedule
- Terminals
- Road network
- Costs (rail, road)

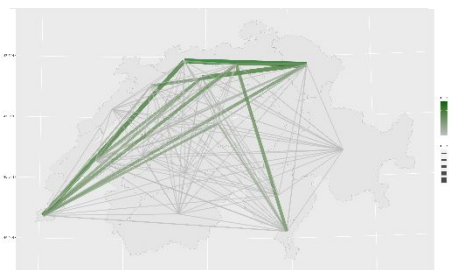


MATSim
Multi-Agent Transport Simulation

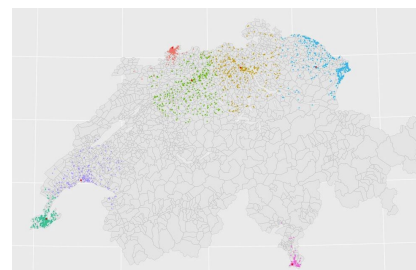


- 1 Agent = 1 Container
- different container types with different time-constraints
- time choice + route choice + mode choice, including terminal choice using the intermodal router
- dynamic simulation of capacity effects (terminals, trains, roads)

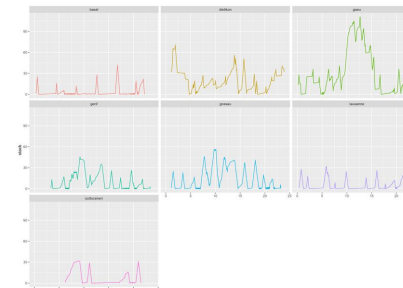
Output



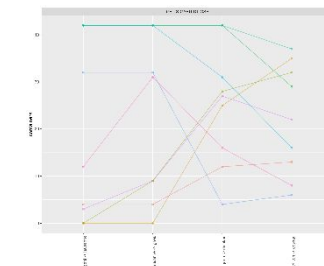
Rail demand per relation



Terminal catchment areas



Terminal loads



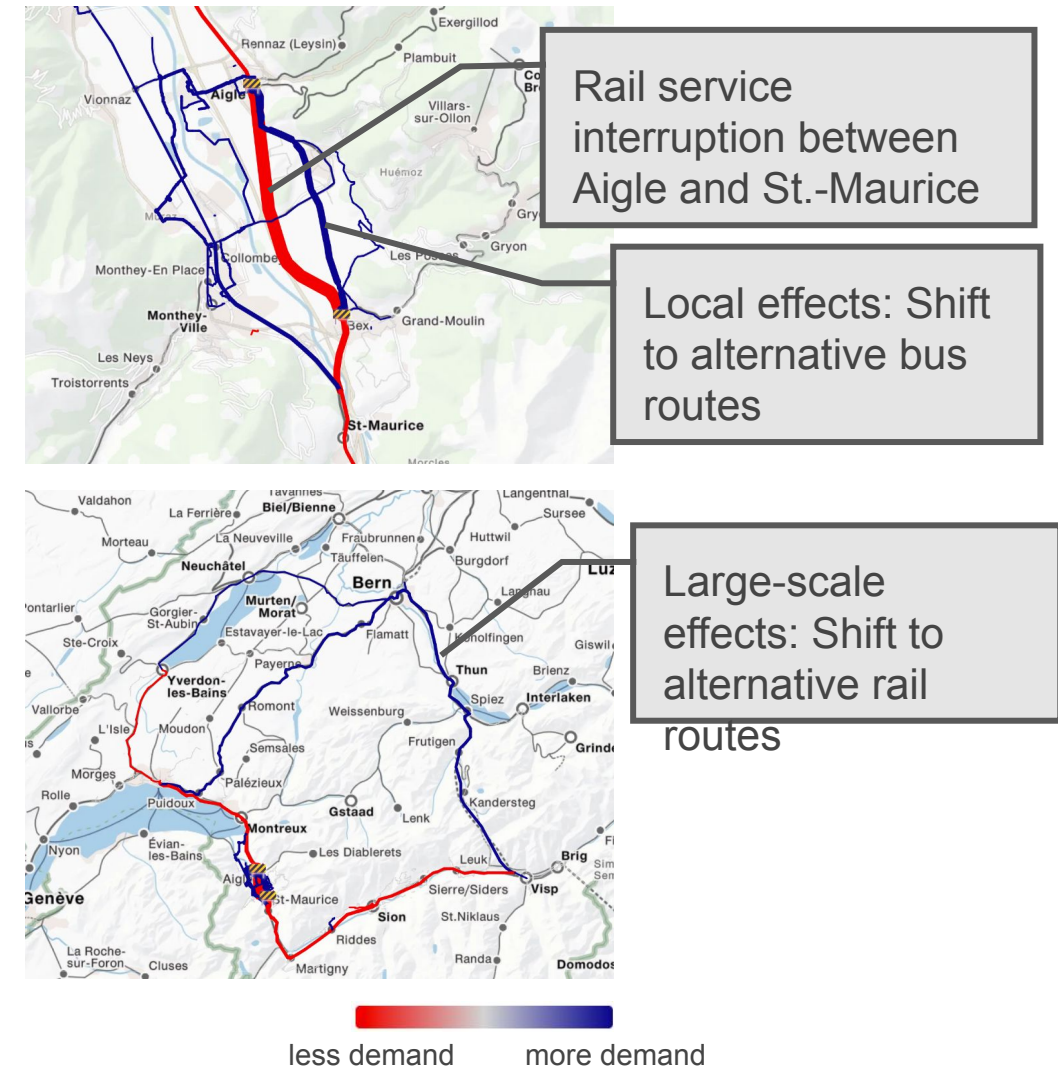
Train loads

MATSim for construction-site planning

Simulating service interruptions

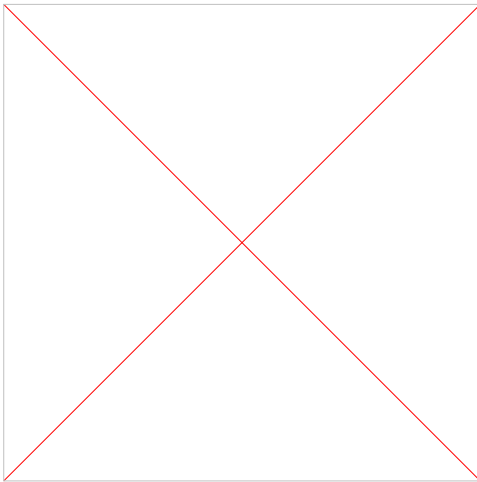
- The rail replacement service (bus) is a relevant share of total construction costs.
- Simulation is used to estimate the demand effects. The demand effects are relevant for the bus service costs.
- Precise knowledge about the costs supports the strategic construction site planning:
 - costs / budgets...
 - nighttime interval construction sites vs. total interruptions with rail replacement service
- We pre-computed service interruptions for all station-station segments in Switzerland.
- Depending on the bus service and the alternative routes, the resulting share rail replacement service users varies from 0% to 100%.

illustrative example



Methodological takeaways

Conclusion



- **Strategic planning at SBB is ongoing.** MATSim will continue to support upcoming project phases and conceptual ideas.
- **Iterative design** and **variants** uncover better solutions.
- **High-resolution simulation** enforces a holistic view and exposes data inconsistencies.
- **AI-assisted development** and **good tools in the MATSim ecosystem** lower the barrier, empowering also non-expert colleagues at SBB to work with MATSim.
- There are still some **challenges**...
 - Random noise and **stochastic effects**: How can we distinguish policy-related demand effects from random fluctuations and other effects unrelated to the policy?
 - Model **calibration**: There will always be KPIs that do not perfectly match the empirical data. The challenge is to achieve a good fit without artificially overfitting the model.
 - Model setup and **configuration**: There is still a significant risk of incorrect simulation settings or configuration, especially as the model becomes more complex.



Thank you.